

Ex. No.: 10a)

Date: 9/4/25

BEST FIT

Aim:

To implement Best Fit memory allocation technique using Python.

Algorithm:

1. Input memory blocks and processes with sizes
2. Initialize all memory blocks as free.
3. Start by picking each process and find the minimum block size that can be assigned to current process
4. If found then assign it to the current process.
5. If not found then leave that process and keep checking the further processes.

Program Code:

```
#include <stdio.h>

int main() {
    int blocksize[10], processsize[10], blockAllocated[10],
        allocation[10];

    int i, j, nb, np;

    printf("Enter number of memory blocks : ");
    scanf("%d", &nb);
    printf("Enter size of each memory block : \n");
    for(i=0; i<nb; i++) {
        printf("Block %d", i+1);
        scanf("%d", &blocksize[i]);
        blockAllocated[i] = 0;
    }

    printf("Enter number of processes : ");
    scanf("%d", &np);
    printf("Enter size of each processes : \n");
    for(i=0; i<np; i++) {
        printf("Process %d", i+1);
```



```
scanf ("%d", & blockSize[i]);
```

```
blockAllocated[i] = 0;
```

```
}  
{
```

```
scanf ("%d", & processSize[i]);
```

```
allocation[i] = -1;
```

```
}  
{
```

```
for(i=0; i < np; i++){
```

```
int bestIdx = -1;
```

```
for(j=0; j < nb; j++){
```

```
if (!blockAllocated[j] & blockSize[j] >= processSize[i]) {
```

```
if (bestIdx == -1 || blockSize[j] <
```

```
blockSize[bestIdx]) {
```

```
bestIdx = j;
```

```
}  
}
```

```
if (bestIdx != -1) {
```

```
allocation[i] = bestIdx;
```

```
blockAllocated[bestIdx] = 1;
```

```
}  
}
```

```
}  
{
```

```
printf ("\n Process No. \t Process Size \t Block No. \n");
```

```
for(i=0; i < np; i++){
```

```
printf ("%d \t \t %d \t \t %d \t \t", i+1,
```

```
processSize[i]);
```

```
}  
}
```

```
}  
}
```

OUTPUT :

The remaining fragments of block :

90
15
13
5
20

Process	Process-size	Block + NO	Fragment
P ₁	20	3	13
P ₂	30	2	15
P ₃	50	5	20
P ₄	40	4	5
P ₅	10	1	90



Sample Output:

Process No.	Process Size	Block no.
1	212	4
2	417	2
3	112	3
4	426	5

Result:

using c program the best fit memory allocation
algorithm is implemented.

[Signature]